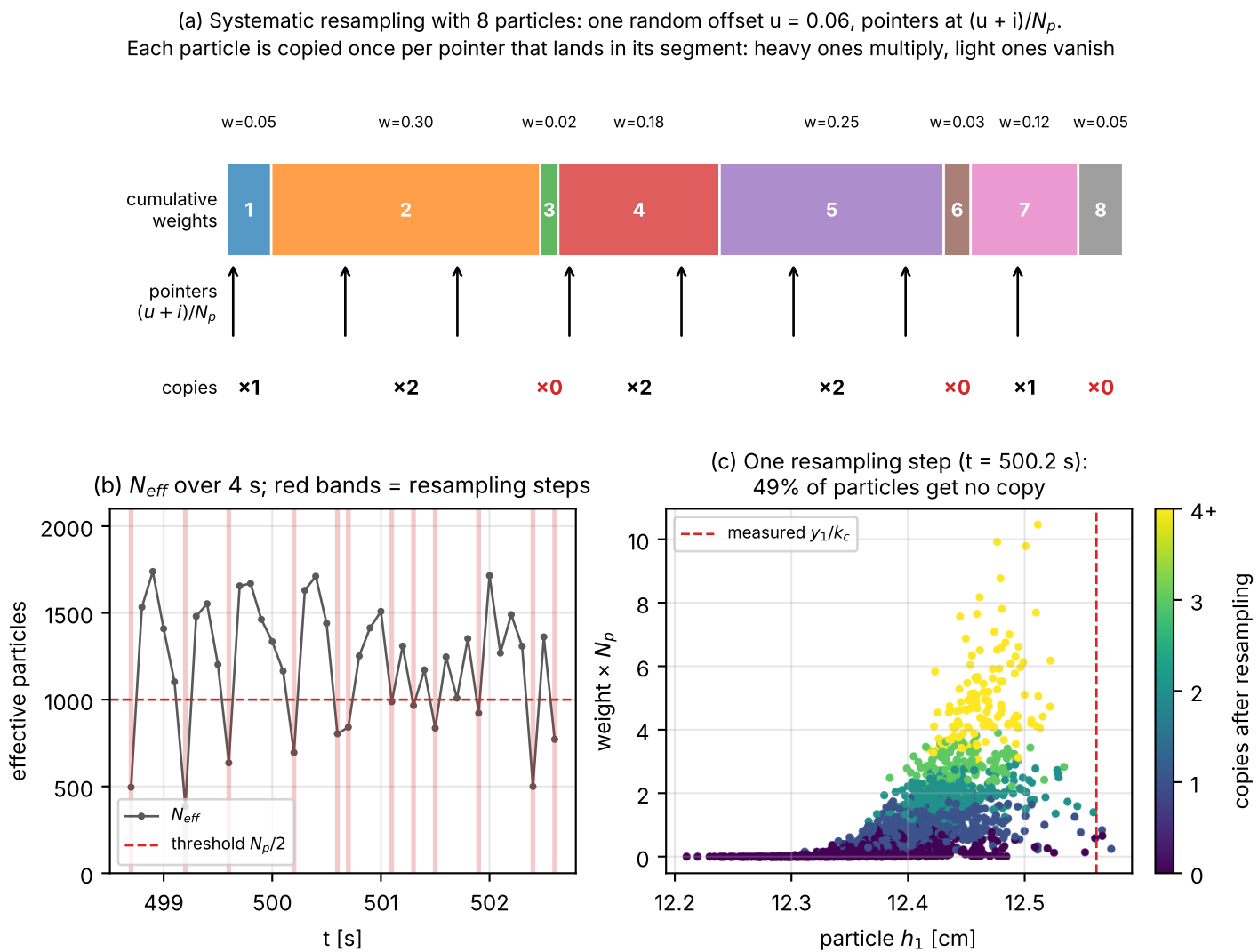

(a) Systematic resampling with 8 particles: one random offset u = 0.06, pointers at (u + i)/N_p.
Each particle is copied once per pointer that lands in its segment: heavy ones multiply, light ones vanish
w=0.05
w=0.30
w=0.02
w=0.18
w=0.25
w=0.03
w=0.12
w=0.05
cumulative weights
1
2
3
4
5
6
7
8
pointers $(u + i)/N_p$
copies
×1
×2
×0
×2
×2
×0
×1
×0
(b) N_{eff} over 4 s; red bands = resampling steps
effective particles
2000
1500
1000
500
0
N_{eff}
threshold $N_p/2$
499
500
501
502
t [s]
(c) One resampling step (t = 500.2 s):
49% of particles get no copy
measured y_1/k_c
weight × N_p
10
8
6
4
2
0
12.2
12.3
12.4
12.5
particle h_1 [cm]
copies after resampling
4+
3
2
1
0